Nathan Senyard

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Education

University of British Columbia — Vancouver, BC
Bachelor of Arts, Dual Major in Computer Science and Philosophy — *Sep. 2020 – May 2026*

Work Experience

New Grad Software Engineer — Sept 2026 – Present
Stripe — *Toronto, Canada*

Machine Learning Research Developer — May 2025 – Aug 2026
UBC Institute for Resources, Environment and Sustainability — *Vancouver, Canada*

- First-authored paper Hedgementation = Hedgerow Segmentation , accepted as at ICLR 2026 Workshop on Machine Learning for Remote Sensing
- Built the end-to-end pipeline behind a 1.2 TB benchmark dataset of Sentinel-2 satellite image time series for hedgerow mapping across France and Canada, covering dataset construction, sampling, concurrent Google Earth Engine export, and verification.
- Designed the lab's shared training framework and dataset for deterministic, reproducible runs of diverse training architectures (e.g., U-TAE) on local lab machines and Alliance Canada GPU clusters via Slurm.

Junior GenAI Developer — May 2025 – Jul 2026
UBC Engineering ChatUBC Project — *Vancouver, Canada*

- Designed and built ChatUBC, a Socratic LLM tutor that coaches engineering students through the ethics of their design projects, running on a five-service Dockerized stack on AWS (FastAPI, vLLM, embedding server, PostgreSQL, web UI) with RAG.
- Served quantized Qwen models with vLLM on a single 24 GB GPU at sub-6s request latency, tuning memory and batching limits for concurrent users.
- Built a scalable asyncio + Redis pipeline that generated 1,000+ synthetic tutoring conversations with Gemini, Claude, and GPT-5 for fine-tuning experiments.

Machine Learning Research Intern — Sept 2024 – Apr 2025
ENEOS Japan — *Yokohama, Japan*

- Accelerated catalyst discovery by developing a custom ensemble screening approach combining TensorFlow neural networks and traditional regression models to screen 132,000+ chemical compounds with 87% test accuracy.
- Rebuilt preprocessing pipelines for 10 GB+ of proprietary chemical data, cutting runtime 40% and saving the team 15+ hours a week.
- Performed feature extraction on raw experimental spectral data, contributing a versatile extraction pipepline and valueable internal dataset.
- Worked entirely in Japanese on a team with no English speakers.

Software Development Research Assistant — Sept 2023 – Sept 2024
University of British Columbia — *Vancouver, BC*

- Fine-tuned GPT-3.5 models for specialized data retrieval achieving 90% accuracy, enabling automated extraction from unstructured academic sources.
- Developed scalable Python/Java web scraper processing 150GB of online data, increasing regression analysis throughput by 300% and supporting large-scale social science research.
- Engineered SQL database handling 200,000+ records with sub-100ms query times.
- Presented findings at IC2S2, a globally recognized computer science conference.

Software Development Coop — Sep. 2022 – Jul 2023
Alida Vancouver — *Vancouver, BC*

- Performed critical audit logging system refactoring, reducing overhead by 70% and decreasing memory usage by 45MB per transaction.
- Optimized SQL queries, cutting average data retrieval times from 1.2s to 0.7s for 125,000+ daily active users.

Awards and Certifications

NSERC Undergraduate Student Research Award(USRA) — May 2026

Japanese Language Proficiency Test(JLPT) N1 — Jul. 2025
157/180 Pts. (96.6th Percentile)